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JOINT ESTIMATION OF PHASE-NOISE, FREQUENCY OFFSET AND CHANNEL USING VARIATIONAL INFERENCE METHODS IN MIMO-OFDM

Abstract

This paper studies the mitigation of phase noise (PHN) and carrier frequency offset (CFO) in MIMO-OFDM (multiple-input multiple-output orthogonal frequency division multiplexing) systems. Further we explore the energy allocation between channel estimation preamble and data symbol in a packet structure. Our mitigation method is based on a set of closed-form iterative equations obtained through using variational inference for data detection and maximum-likelihood estimates for channel estimation. We extend the work of these methods for SISO-OFDM systems to the case of MIMO-OFDM.

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Glossary

BER	Bit Error Rate
CP	Cyclic Prefix
CPE	Common Phase Error
CRLB	Cràmer-Rao Lower Bound
DFT	Discrete Fourier Transform
FFT	Fast Fourier Transform
ICI	Inter-carrier Interference
IDFT	Inverse Discrete Fourier Transform
IFFT	Inverse Fast Fourier Transform
ISI	Inter-symbol Interference
LS	Least Square
MIMO	Multiple-Input Multiple-Output
MMSE	Minimum Mean Square Error
MSE	Mean Square Error
OFDM	Orthogonal Frequency Division Multiplexing
PHN	Phase Noise
PLL	Phase-locked Loop
QoS	Quality of Service
SINR	Signal-to-Interference and Noise Ratio
SISO	Single-Input Single-Output
SVD	Singular Value Decomposition
VCO	Voltage-Controlled Oscillator
WSS	Wide Sense Stationary

I. Introduction

THE GROWING demand for wireless multimedia services has led to the need for high-speed and reliable communication with high spectral efficiency. However these wireless communication channels are subjected to multipath fading, especially frequency-selective fading which impact the speed and reliability of data transmission. In light of these obstacles, MIMO-OFDM (Multiple Input Multiple Output – Orthogonal Frequency Division Multiplexing) offers natural synergies to combat these impairments and has emerged as the leading technology for implementation and deployment in future wireless systems[1].

A. MIMO-OFDM

In OFDM, a wideband high speed data stream is divided into multiple flat low-rate narrowband sub-channels or sub-carriers via the Inverse Discrete Fourier Transform (IDFT) such that multiple symbols can be transmitted in parallel. This has several advantages: 1) resistance against the effects of frequency-selective fading, 2) reduced equalizer complexity, and 3) efficient use of the spectrum via overlapping in the frequency domain. The original data symbols are obtained at the receiver by applying the DFT to the received symbols. Inter-symbol Interference (ISI) is eliminated and circular convolution achieved through appending a Cyclic Prefix (CP) to the front of the OFDM symbol. Finally, OFDM allows for flexible allocation of resources and diverse functionalities on one radio link as each subcarrier can be made to carry different media (video, graphics, speech, text etc.) with different modulation schemes depending on Quality of Service (QoS) requirements or offer multiple access schemes. The ease with which OFDM can be implemented physically using Fast Fourier Transform (FFT) and Inverse Fast Fourier Transform (IFFT) along with the advantages described above has lead it to be advocated as the ideal air interface for many current and future standards such as Digital Audio Broadcasting (DAB), Digital Video Broadcasting for Terrestrial Television (DVB-T), Wireless Local Area Networks (WLANs), Broadband Radio Access Networks (BRANs), IEEE 802.11a/g/n, LTE and WiMAX. On the other hand, MIMO is a class of system that uses multiple transmit and receive antennas. The use of multiple antennas allows for the multiplexing of signals leading to a higher data rate proportional to the number of transmitters and receivers without the need for extra bandwidth or power and offer greater reliability under highly challenging channel environments.

B. Research Gap, Objectives and Significance

While MIMO-OFDM provides great flexibility, low error rates and high spectral efficiency, it places strict demands on orthogonality of the OFDM subcarriers and correct channel side information at the receiver and transmitter. However, imperfect oscillators and Doppler shifts cause phase shifts and carrier frequency offsets that has three deleterious effects on a MIMO-OFDM system: (1) a common phase error (CPE) resulting in the rotation of the constellation set[2][3], (2) inter-symbol interference (ICI) due to destruction of the orthogonality between OFDM sub-carriers[2][3] and (3) interference between transmitters and SNR degradation due to imperfect channel estimation.

Numerous papers have been written on the topic of phase noise suppression, carrier frequency synchronization and channel estimation in OFDM and MIMO-OFDM systems as will be described in section II. However, none of them have visited the problem from a holistic perspective involving all three parameter and none have looked at the signal-to-noise ratio (SNR) interdependencies between the channel estimation and data detection that arise when symbols are not viewed as distinct individuals but in a packet structure.

Therefore, we will first find a way to integrate the various works done in this area into a functional MIMO-OFDM system capable of joint channel estimation under the influence of phase noise and carrier frequency offset. This is then followed by joint data estimation with the obtained channel estimate. We hope to achieve this through the use of variational inference algorithms developed in [4] and [5] for SISO-OFDM and extending it to the case of MIMO-OFDM implemented using Singular Value Decomposition (SVD). Further we would like to explore the optimum power allocation between training and data symbols in a MIMO-OFDM packet under such a scenario.

The significance of our work is that it will allow future wireless systems which will implement OFDM with MIMO architecture to correct for multiplicative time-varying effects for uplink and downlink, leading to improved system performance and crucial resistance to oscillator impairments.

II. Literature Review

MIMO-OFDM systems are extremely sensitive to time-varying multiplicative effects of fast fading, Doppler shifts and oscillator jitter. Under fast fading conditions, OFDM systems cannot be implemented effectively. Luckily, most channels in urban settings where the benefits of MIMO-OFDM are most discernible do not suffer from this detriment. However the latter two effects introduce a mismatch between the carrier frequencies of the received signal and the local oscillator that must be corrected both for channel estimation and later data detection. Further, oscillator jitter introduces a very damaging effect called phase noise (PHN) where the phase of the generated sinusoid of the local oscillator changes randomly over time.

A. Overview

While channel estimation and carrier frequency offsets remain relatively constant over many OFDM symbols and therefore training symbol estimates can be used in subsequent data detection, phase noise varies randomly from subcarrier to subcarrier on a per symbol basis. Phase noise arising from imperfections in the oscillator has been extensively studied in [2], [6]-[15]. In [2], Robertson and Kaiser analyzed the influence of the number of subcarriers and power spectral density output at the oscillator on the statistics of both CPE and ICI. They proposed a pilot-based method of estimating the common phase error through averaging the phase shift on each pilot symbol. A more general system model for phase noise in OFDM systems was given in [7], where a closed form equation was found for SINR degradation for given phase-noise line-width, transmission rate, number of subcarriers and SNR. A normalized MMSE estimate was used to compensate for the ICI along with pilot-aided common phase error estimate. This was extended in [8] by the use of decision feedback and applied to WLAN systems implementing OFDM and [6] where it was adapted to MIMO-OFDM systems implemented with V-BLAST. Analysis of phase noise in MIMO-OFDM systems were performed in [9] and [11] where a pilot-based least square (LS) method was proposed to combat the effects of CPE. The power of ICI component as function of the -3dB bandwidth of oscillator output was analyzed in [10]. However, all of the above assumes perfect channel estimation and synchronization which in reality would suffer from PHN themselves and either ignores ICI or treats it as an extraneous noise term. A Kalman Filter method for estimation of CPE in OFDM was proposed in [12] and [13]. Finally, in [14] and [15], a method was

proposed to estimate higher orders of PHN rather than just the dc CPE component. However this suffers from error propagation issues that rely greatly on getting the initial data estimates and assumes that the energy of ICI is concentrated in lower part of the frequency domain.

Unlike phase noise, channels and carrier frequency offset vary slowly and can be assumed to be quasi-static over several OFDM symbol times. In [3], Moose showed that two repetitive OFDM sequences can be used to estimate carrier frequency offset. This was improved upon in [17] where Schmidl showed that both coarse timing synchronization and frequency synchronization can be performed using one OFDM symbol by sending nulls on odd subcarriers. Both cases were implemented for OFDM systems only and did not factor in phase noise or deal with channel estimation. A method was proposed in [16] for frequency synchronization in MIMO-OFDM systems, however it required Q OFDM symbols for Q transmitters in order to obtain a unique solution and did not factor in the effects of phase noise. In [18-20], different training-symbol sequences were proposed. In [18], comb-like training sequence was proposed for channel estimation in a 2x2 OFDM system. However the method proposed performed channel estimation in the frequency domain which is suboptimal as most channels have less taps than subcarriers. It was shown in [19-20] that the optimum channel estimation training symbols must be equal-powered, equal-spaced, orthogonal and shift-orthogonal. A more complete approach was taken in [21] and [22]. In [21], channel estimation MSE was extended with phase noise statistics which was subsequently used in data detection where it was shown that frequency or code orthogonal sequences performed better than time-orthogonal sequences for channel estimation and data detection under the influence of phase noise. In [22], a LS estimator for phase noise and channel estimation in MIMO systems was shown and the Cràmer-Rao lower bounds were derived on the channel estimate. A complete system was considered in [23] where a V-BLAST MIMO-OFDM WLAN was implemented using time-orthogonal training sequence for frequency synchronization and LS channel estimate and pilot symbols were used for common phase error correction.

A great variety of schemes have been proposed in the area of phase noise, frequency offset and channel estimation. A more general approach was taken in [4] and [5] where a closed form maximum likelihood function was derived for joint estimation of all the concerned system parameters. In [4], it presented an algorithm that approximated the complex joint phase noise and data distributions to simpler independent continuous Gaussian distributions. From this theoretical concept, the minimization of the

Kullback-Leibler divergence between the distributions became the same as the maximization of the MLE (Maximum Likelihood Estimator) for the former complex distribution. This allowed the development of a set of iterative equations which jointly estimated data and phase noise. It showed that not only is the estimation of MLE for both ICI and CPE possible, but it is of lower complexity than previously believed. In a similar manner, a joint estimate for phase noise, carrier frequency offset and channel were shown in [5].

In this thesis, we revisit the work done in [4],[5] and in combination with [17],[19-20] implement a complete system similar in scope to [23] but extending the prior work to the case of SVD MIMO-OFDM as SVD provides a more straightforward means of scaling to multi-user case than V-BLAST. We will show in section III how to jointly estimate CFO, PHN and the MIMO-OFDM channel with optimum training sequences which will then be used with variational methods for data detection. Our results will be shown in section IV with conclusion in section V.

B. Mathematic Background

The relevant mathematic background will be reproduced here in condensed form for the purpose of completeness and allow for a more coherent argument in section III. For more detailed proofs and in-depth understanding, the reader may refer to [4], [5], [17], [19], [24].

In the following sections, the following notation will be used: upper and lower case bold face letters will indicate matrices and column vectors while non-bolded lower case letters will indicate scalars; $(\cdot)^T$, $(\cdot)^*$, $(\cdot)^+$ and $(\cdot)^H$ denote transpose, conjugation, pseudo-inverse and Hermitian transpose; $N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ and $CN(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ represent real and circularly symmetric complex Gaussian with mean $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$; and $\hat{\boldsymbol{x}}$ is an estimate of $\boldsymbol{x}$.

B.1 Phase Noise

Two statistical models exist for phase noise[4]. The first models PHN as non-stationary Wiener process with its variance growing with time (Wiener PHN). The second models PHN as a zero-mean coloured Gaussian process that is wide-sense stationary (WSS) with finite variance (Gaussian PHN). In both cases, if we assume $\theta(t)$ as the PHN output of the voltage-controlled oscillator (VCO), then for the

m th OFDM symbol – $p(\boldsymbol{\theta}_m) = N(\mathbf{0}, \boldsymbol{\Phi})$, where the samples are taken at rate of N_c/T samples per second – N_c is the number of OFDM subcarriers and T is the period of an OFDM symbol.

1) For Wiener PHN, the discrete samples of $\theta(t)$ form a random-walk process:

$$\begin{aligned}\theta_n &= \theta_{n-1} + \varphi_n \\ \text{where } n &= 0, \dots, N_c - 1 \\ p(\varphi_n) &= N(0, \alpha_\varphi^2) \\ \alpha_\varphi^2 &= 4\pi\Delta f_{3dB}T \quad [12]\end{aligned}\tag{1}$$

If we assume perfect synchronization at the beginning of each OFDM symbol, then $\theta_{-1} = 0$. The Gaussian-distributed PHN vector $\boldsymbol{\theta} = [\theta_0, \dots, \theta_{N_c-1}]^T$ would thus have the covariance matrix:

$$\boldsymbol{\Phi} = \alpha_\varphi^2 \begin{bmatrix} 1 & 1 & \dots & 1 \\ 1 & 2 & \dots & 2 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 2 & \dots & N_c \end{bmatrix}\tag{2}$$

2) For Gaussian PHN, we can write the output of the VCO with PHN as:

$$s(t) = e^{j(2\pi f_c t + \theta(t))}\tag{3}$$

Then the autocorrelation function of $s(t)$, $R_s(\tau)$, can be calculated as:

$$\begin{aligned}R_s(\tau) &= E\{s^*(t)s(t+\tau)\} = e^{j2\pi f_c \tau} E\{e^{j(\theta(t+\tau)-\theta(t))}\} \\ &= e^{j2\pi f_c \tau} e^{-R_\theta(0)} e^{-R_\theta(\tau)} \approx e^{j2\pi f_c \tau} e^{-R_\theta(0)} (1 + R_\theta(\tau))\end{aligned}\tag{4}$$

If we were to then take the Fourier transform, then the PSD of $s(t)$ becomes,

$$S_s(f) = e^{-R_\theta(0)} (\delta(f - f_c) + S_\theta(f - f_c))\tag{5}$$

Since the shape of $S_s(f)$ can be measured by a spectrum analyzer or provided in the VCO specifications, the $S_\theta(f)$ and by extension $R_\theta(\tau)$ can be found. The autocorrelation is also the autovariance function as the PHN process is zero mean, in which case the i th and j th column of Φ becomes:

$$\Phi_{i,j} = R_\theta\left(|i-j|\frac{T}{N}\right) \quad (6)$$

In [5], the sensitivity to inexact knowledge of Φ is further examined.

B.2 Carrier Frequency Offset

The statistics of CFO is relatively simple in comparison. Assume Δf_c be half of the subcarrier bandwidth, then on each OFDM packet, the CFO is modeled as a uniform distribution between $[-\Delta f_c, \Delta f_c]$.

B.3 Joint PHN, CFO and Channel Estimation in SISO-OFDM

In the OFDM packet structure shown in Figure 1. Channel estimation and frequency synchronization is usually performed in a block structure using a short preamble training symbol. Below a shortened derivation for joint estimation of PHN, CFO and channel from [5] will be presented.

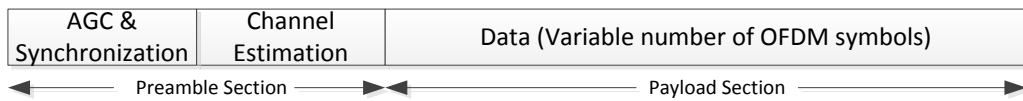
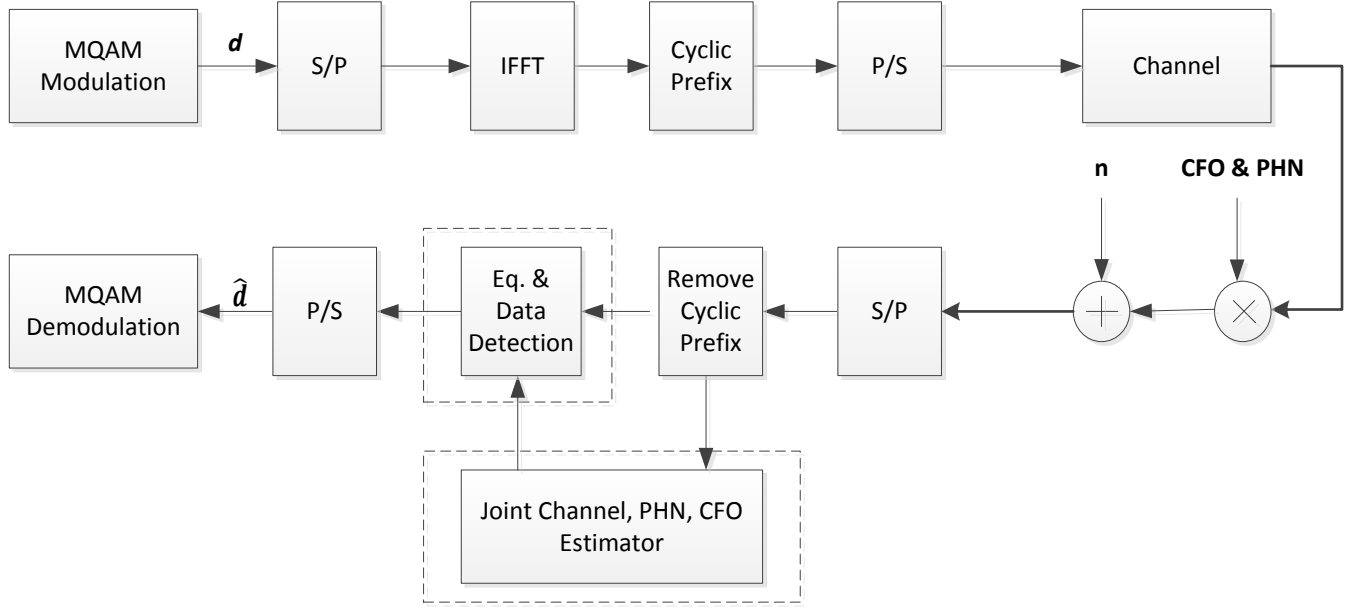


Figure 1. SISO-OFDM Packet Structure

Assuming perfect timing synchronization and transceiver structure shown in Figure 2, the complex baseband received signal for SISO-OFDM with transmissions only on even subcarriers can be represented as

$$r_n = \frac{1}{\sqrt{N}} e^{j(\theta_n + \frac{2\pi\epsilon n}{N})} \sum_{k=0}^{N_c/2-1} h_k d_k e^{j4\pi n k / N_c} + \eta_n \quad \text{for } n = 0, \dots, N_c - 1 \quad (7)$$



where $\epsilon = \Delta f_c T$; $\{\theta_n\}_{n=0}^{N_c-1}$ is the discrete-time PHN sequence; $\{h_k\}_{k=0}^{N_c-1}$ is the channel frequency response; $\{d_k\}_{k=0}^{N_c-1}$ is the transmitted symbols belonging to an MQAM constellation; and $\{\eta_n\}_{n=0}^{N_c-1}$ is the complex Gaussian noise with variance $2\sigma_n^2$. Equation (7) can be rewritten in matrix form as

$$\mathbf{r} = \mathbf{E}\mathbf{P}\check{\mathbf{F}}^H \mathbf{D}\mathbf{W}\mathbf{g} + \mathbf{n} \quad (8)$$

where $\mathbf{E} = \text{diag}\left(\left[1, e^{\frac{j2\pi\epsilon}{N_c}}, \dots, e^{\frac{j2\pi(N_c-1)\epsilon}{N_c}}\right]^T\right)$; $\mathbf{P} = \text{diag}([1, e^{j\theta_0}, \dots, e^{j\theta_{N_c-1}}]^T)$; $\check{\mathbf{F}} = [\mathbf{F}, \mathbf{F}] \in \mathbb{C}^{\frac{N_c}{2} \times N_c}$

is the stacked DFT matrix; $\mathbf{D} = \text{diag}([d_0, d_2, \dots, d_{N_c-2}]^T)$; $\mathbf{W} \in \mathbb{C}^{\frac{N_c}{2} \times L}$ is the first L columns and even rows of the full DFT matrix; $\mathbf{n}$ is the noise vector; and $\mathbf{g} = [g_0, \dots, g_{L-1}]^T$ is the L tap channel in the time domain. Because the transmitted OFDM symbol has two identical halves due to transmission on only even subcarriers, we can rewrite (8) where $\mathbf{r}_1 = [r_0, \dots, r_{N_c/2-1}]^T$ and $\mathbf{r}_2 = [r_{N_c/2}, \dots, r_{N_c-1}]^T$ as

$$\mathbf{r}_1 = \mathbf{P}_1 \mathbf{x} + \mathbf{n}_1 \quad \text{and} \quad \mathbf{r}_2 = e^{j\pi\epsilon} \mathbf{P}_2 \mathbf{x} + \mathbf{n}_2 \quad (9)$$

where $\mathbf{x} = \mathbf{E}\mathbf{G}\mathbf{F}^H \mathbf{D}\mathbf{W}\mathbf{g}$. All terms in $\mathbf{x}$ are as before but only half the size. Equation (9) can be re-expressed as

$$\mathbf{r}_2 = e^{j\pi\epsilon} \mathbf{P}_\Delta \mathbf{r}_1 - e^{j\pi\epsilon} \mathbf{P}_\Delta \mathbf{n}_1 + \mathbf{n}_2 \quad \text{where, } \mathbf{P}_\Delta = \mathbf{P}_2 \mathbf{P}_1^H \quad (10)$$

The covariance matrix of $\mathbf{P}_\Delta$ and $p(\boldsymbol{\theta}_\Delta = \boldsymbol{\theta}_2 - \boldsymbol{\theta}_1)$ can be expressed as

Figure 2. SISO-OFDM System Model

$$\boldsymbol{\Phi} = \begin{bmatrix} \boldsymbol{\Phi}_{N/2} & \boldsymbol{\Upsilon} \\ \boldsymbol{\Upsilon} & \boldsymbol{\Phi}_{N/2} \end{bmatrix} \rightarrow \boldsymbol{\Phi}_\Delta = 2\boldsymbol{\Phi}_{N/2} - \boldsymbol{\Upsilon} - \boldsymbol{\Upsilon}^T \rightarrow p(\boldsymbol{\theta}_\Delta) = N(\mathbf{0}, \boldsymbol{\Phi}_\Delta) \quad (11)$$

then if $\mathbf{R}_1 = \text{diag}(\mathbf{r}_1)$, the optimal estimate ϵ during the synchronization stage of the SISO-OFDM packet becomes

$$\begin{aligned} \hat{\epsilon} &= \arg \max_{\epsilon} p(\mathbf{r}_1, \mathbf{r}_2 | \epsilon) = \arg \max_{\epsilon} p(\mathbf{r}_2 | \mathbf{r}_1, \epsilon) = \arg \min_{\epsilon} \log p(\mathbf{r}_2 | \mathbf{r}_1, \epsilon) \\ &= \arg \min_{\epsilon} \log \text{CN}(e^{j\pi\epsilon} \mathbf{r}_1, \mathbf{R}_1 \boldsymbol{\Phi}_\Delta \mathbf{R}_1^H + 4\sigma_n^2 \mathbf{I}) \\ &= \frac{1}{\pi} \nabla \mathbf{r}_1^H (\mathbf{R}_1 \boldsymbol{\Phi}_\Delta \mathbf{R}_1^H + 4\sigma_n^2 \mathbf{I})^{-1} \mathbf{r}_2 \end{aligned} \quad (12)$$

with the optimal estimate of ϵ , we can then estimate the optimal joint estimator for phase noise and channel as

$$p(\mathbf{r}, \boldsymbol{\theta}, \mathbf{g}) = p(\mathbf{r} | \boldsymbol{\theta}, \mathbf{g}) p(\boldsymbol{\theta}) \quad (13)$$

Where the negative log-likelihood function can be expressed as

$$\begin{aligned} L(\boldsymbol{\theta}, \mathbf{g}) &= -\log p(\mathbf{r} | \boldsymbol{\theta}, \mathbf{g}) - \log p(\boldsymbol{\theta}) \\ &= \frac{1}{2\sigma_n^2} (\mathbf{r} - \hat{\mathbf{E}} \mathbf{P} \check{\mathbf{F}}^H \mathbf{D} \mathbf{W} \mathbf{g})^H (\mathbf{r} - \hat{\mathbf{E}} \mathbf{P} \check{\mathbf{F}}^H \mathbf{D} \mathbf{W} \mathbf{g}) + \frac{1}{2} \boldsymbol{\theta}^T \boldsymbol{\Phi}^{-1} \boldsymbol{\theta} \end{aligned} \quad (14)$$

Maximizing the likelihood function for $\boldsymbol{\theta}$ and $\mathbf{g}$,

$$\hat{\boldsymbol{\theta}} = [\text{Re}(\hat{\mathbf{E}} \mathbf{A} \hat{\mathbf{E}}^H) + 4\sigma_n^2 \rho^2 \boldsymbol{\Phi}^{-1}]^{-1} \text{Im}(\hat{\mathbf{E}} \mathbf{A} \hat{\mathbf{E}}^H) \mathbf{1} \quad (15)$$

$$\hat{\mathbf{g}} = (4\rho^2)^{-1} \mathbf{W}^H \mathbf{D}^H \tilde{\mathbf{F}} \mathbf{P}^H \hat{\mathbf{E}}^H \mathbf{r} \quad (16)$$

Where $\mathbf{A} = \mathbf{R}^H \begin{bmatrix} \mathbf{F}^H \mathbf{D} & \mathbf{0} \\ \mathbf{0} & \mathbf{F}^H \mathbf{D} \end{bmatrix} \begin{bmatrix} \mathbf{I} + \mathbf{V}\mathbf{V}^H & -\mathbf{W}\mathbf{W}^H \\ -\mathbf{W}\mathbf{W}^H & \mathbf{I} + \mathbf{V}\mathbf{V}^H \end{bmatrix} \begin{bmatrix} \mathbf{F}^H \mathbf{D} & \mathbf{0} \\ \mathbf{0} & \mathbf{F}^H \mathbf{D} \end{bmatrix}^H \mathbf{R}$; $2\rho^2$ is the average symbol energy; and $\mathbf{F} = [\mathbf{W}|\mathbf{V}]$ where $\mathbf{W} \in \frac{N_c}{2} \times L$ and $\mathbf{V} \in \frac{N_c}{2} \times (\frac{N_c}{2} - L)$.

B.4 Variational Methods for SISO-OFDM Data Detection

In this step, the same model as Figure 2 is again used in [4]. However the unknowns here are the data symbols and PHN sequence. The system model is similar to (8) with the distinct differences that perfect frequency synchronization is assumed and data symbols are non-null for all subcarriers. The system model is thus shown in equation (17) where the second equality arises from small phase angle assumption.

$$\mathbf{r} = \mathbf{P}\mathbf{F}^H \mathbf{H} \mathbf{d} + \mathbf{n} = \text{diag}(\mathbf{F}^H \mathbf{H} \mathbf{d})(\mathbf{1} + j\boldsymbol{\theta}) + \mathbf{n} \quad (17)$$

Given the above, we thus have the following prior distributions,

$$\begin{aligned} p(\mathbf{d}) &= \text{CN}(\mathbf{0}, 2\rho^2 \mathbf{I}) \\ p(\boldsymbol{\theta}) &= N(\mathbf{0}, \boldsymbol{\Phi}) \\ p(\mathbf{r}|\mathbf{d}, \boldsymbol{\theta}) &= \text{CN}(\text{diag}(\mathbf{F}^H \mathbf{H} \mathbf{d})(\mathbf{1} + j\boldsymbol{\theta}), 2\sigma_n^2 \mathbf{I}) \end{aligned} \quad (18)$$

Integrating out $\boldsymbol{\theta}$ as a nuisance parameter, we obtain

$$p(\mathbf{r}|\mathbf{d}) = \text{CN}(\mathbf{F}^H \mathbf{H} \mathbf{d}, \text{diag}(\mathbf{F}^H \mathbf{H} \mathbf{d}) \boldsymbol{\Phi} \text{diag}(\mathbf{F}^H \mathbf{H} \mathbf{d})^H + 2\sigma_n^2 \mathbf{I}) \quad (19)$$

However the above problem is intractable as there is no closed-form solution to this problem and has complexity of $O(M^N)$ where M is constellation size and N the number of subcarriers. Fortunately, through applying variational inference, we can find a closed form solution to (19). Re-expressing (19) as

$$p(\mathbf{r}|\mathbf{d}) = \int p(\mathbf{d}, \boldsymbol{\theta}|\mathbf{r}) d\boldsymbol{\theta} \quad (20)$$

We can approximate the distribution of $p(\mathbf{d}, \boldsymbol{\theta}|\mathbf{r})$ with a hypothetical distribution $Q(\mathbf{d}, \boldsymbol{\theta})$. Further we can assume that $Q(\mathbf{d}, \boldsymbol{\theta}) = Q(\mathbf{d})Q(\boldsymbol{\theta})$ (via *mean-field* approximation). In this manner, the strength of the variational inference approach can be realized. If the hypothetical distributions are properly selected, the problem becomes finding a $\mathbf{d}$ and $\boldsymbol{\theta}$ that maximizes the distributions $Q(\mathbf{d})$ and $Q(\boldsymbol{\theta})$ that most closely matches $p(\mathbf{d}, \boldsymbol{\theta}|\mathbf{r})$ instead of maximizing $\mathbf{d}$ and $\boldsymbol{\theta}$ over $\int Q(\mathbf{d}, \boldsymbol{\theta}) d\boldsymbol{\theta}$ which might prove again infeasible. For example, if we assume that $Q(\mathbf{d}) = \delta(\mathbf{d}, \hat{\mathbf{d}})$ and $Q(\boldsymbol{\theta}) = \delta(\boldsymbol{\theta}, \hat{\boldsymbol{\theta}})$ or otherwise known as delta functions, the Kullback-Leibler divergence between $Q(\mathbf{d}, \boldsymbol{\theta})$ and $p(\mathbf{d}, \boldsymbol{\theta}|\mathbf{r})$ can be expressed as

$$F(Q, p) = \int Q(\mathbf{d})Q(\boldsymbol{\theta}) \log \frac{Q(\mathbf{d})Q(\boldsymbol{\theta})}{p(\mathbf{r}|\mathbf{d}, \boldsymbol{\theta})p(\mathbf{d})p(\boldsymbol{\theta})} d\mathbf{d}d\boldsymbol{\theta} \quad (21)$$

Solving for this equation means finding the $Q(\mathbf{d})$ and $Q(\boldsymbol{\theta})$ that most closely resembles $p(\mathbf{d}, \boldsymbol{\theta}|\mathbf{r})$. Since we chose $Q(\mathbf{d}) = \delta(\mathbf{d}, \hat{\mathbf{d}})$ and $Q(\boldsymbol{\theta}) = \delta(\boldsymbol{\theta}, \hat{\boldsymbol{\theta}})$, $\hat{\mathbf{d}}$ and $\hat{\boldsymbol{\theta}}$ automatically becomes the optimal estimates. Therefore applying the matrix properties in [24] to (21), a set of equations was obtained that iteratively estimates $\boldsymbol{\theta}$ and $\mathbf{d}$

$$\hat{\boldsymbol{\theta}}^{(t)} = [\sigma_n^2 \boldsymbol{\Phi}^{-1} + \mathbf{X}^H \mathbf{X}]^{-1} \text{Im}(\mathbf{X}^H (\mathbf{r} - \mathbf{F}^H \mathbf{H} \hat{\mathbf{d}}^{(t-1)})) \quad (22)$$

$$\hat{\mathbf{d}}^{(t)} = [\sigma_n^2 \rho^{-2} \mathbf{I} + \mathbf{H}^H \mathbf{H}]^{-1} \mathbf{H}^H \mathbf{F} \text{diag}(e^{j\hat{\boldsymbol{\theta}}^{(t)}}) \mathbf{r} \quad (23)$$

where $\mathbf{X} = \text{diag}(\mathbf{F}^H \mathbf{H} \hat{\mathbf{d}}^{(t-1)})$. The matrix properties and more detailed derivation used to find (21) and (22) can be found in [4].

B.5 Optimum Training Symbol for MIMO-OFDM

We will show an abridged derivation of the optimal training symbols in [19] as background for our analysis in section III. Let $\mathbf{h}^{q,r}$ be the L tap channel response between the r th transmitter and q th receiver. If $\mathbf{b}^r$ be the vector of transmitted non-zero training symbols, then we have the following equation in the frequency domain

$$\mathbf{y}^q = \sum_{r=1}^{N_t} \text{diag}(\mathbf{b}^r) \mathbf{F} \mathbf{h}^{q,r} + \boldsymbol{\eta}^q$$

$$\mathbf{y}^q = \mathbf{A} \mathbf{h}^q + \boldsymbol{\eta}^q$$

where,

$$\begin{aligned}\mathbf{A} &= [\mathbf{b}^1 \mathbf{F}, \dots, \mathbf{b}^{N_t} \mathbf{F}] \\ \mathbf{h}^q &= [\mathbf{h}^{q,1^T}, \dots, \mathbf{h}^{q,N_t^T}]\end{aligned}\quad (24)$$

Using a LS estimator,

$$\hat{\mathbf{h}}^q = \mathbf{A}^+ \mathbf{Y}^q = \mathbf{h}^q + \mathbf{A}^+ \boldsymbol{\eta}^q \quad (25)$$

Then the MSE of the LS channel estimate becomes

$$MSE = \frac{1}{LN_t} E\{|\hat{\mathbf{h}}^q - \mathbf{h}^q|\} = \frac{\sigma_n^2}{LN_t} \text{tr}\{(\mathbf{A}^H \mathbf{A})^{-1}\} \quad (26)$$

Therefore, in order to minimize the mean squared error, we require that $\mathbf{A}^H \mathbf{A}$ to be diagonal. If we re-write the $LN_t \times LN_t$ matrix $\mathbf{A}^H \mathbf{A}$ in the form of $L \times L$ sub-matrices $\mathbf{C}_{r,s}$

$$\mathbf{A}^H \mathbf{A} = \begin{bmatrix} \mathbf{C}_{1,1} & \cdots & \mathbf{C}_{1,N_t} \\ \vdots & \ddots & \vdots \\ \mathbf{C}_{N_t,1} & \cdots & \mathbf{C}_{N_t,N_t} \end{bmatrix} \quad (27)$$

Assuming that each subcarrier has power $\mathcal{E}$, then the following conditions must be satisfied by $\mathbf{C}_{r,s}$

$$\mathbf{C}_{r,s} = \begin{cases} \mathcal{E} \mathbf{I}_L & , \text{ if } r = s \\ 0_{L \times L} & , \text{ if } r \neq s \end{cases} \quad (28)$$

As shown in [19], as long as the number of pilot symbols $> LN_t$, then the first part of (28) is satisfied when 1) each non-zero subcarrier p is equal-powered and 2) the subcarriers are equal-spaced. The second equality is satisfied for channels with $L > 1$ when the pilot sequences on each antenna are mutually phase-shift orthogonal over the range of $e^{-j2\pi\varphi/P}$ where φ is over the range of $-L + 1$ to $L - 1$ or in other words, the pilot sequences form a circulant matrix in the time-domain. With these conditions, a possible set of optimal pilot symbols is

$$\begin{aligned}
 b_p^r &= \sqrt{\epsilon} e^{-j2\pi n_r p/P} \quad \text{where } n_r = (r-1)L \\
 \forall p &\in \{0, \dots, P-1\} \\
 \forall r &\in \{1, \dots, N_t\}
 \end{aligned} \tag{29}$$

which is a simple rotation of the same set of pilot symbols around the imaginary-real plane spaced out at intervals determined by n_r .

B.6 SVD MIMO

Consider a MIMO system channel $\mathbf{H}$, where the $h_{q,r}$ entry is the frequency response of the channel between transmitter q and receiver r . We can rewrite the channel $\mathbf{H} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^H$ where $\mathbf{U}$ and $\mathbf{V}$ are unitary matrices and $\mathbf{\Sigma}$ is a diagonal vector consisting of the singular values of $\mathbf{H}$. If we have

$$\mathbf{y} = \mathbf{H}\mathbf{d} + \mathbf{n} \tag{30}$$

then we can break down $\mathbf{H}$ into parallel SISO channels by multiplying $\mathbf{d}$ by $\mathbf{V}$ at the transmitter and $\mathbf{U}^H$ at the receiver

$$\mathbf{y} = \mathbf{U}^H \mathbf{H} \mathbf{V} \mathbf{d} + \mathbf{U}^H \mathbf{n} = \mathbf{\Sigma} \mathbf{d} + \mathbf{U}^H \mathbf{n} \tag{31}$$

The advantage of this method is that it allows us to flexibly and efficiently exploit the multiple channels of $\mathbf{H}$ for both diversity and capacity gain through selecting the desired matrices $\mathbf{V}$ and $\mathbf{U}$.

III. Methods

IN THE following parts, we will describe the extension of the work described in Section II to MIMO-OFDM. Section III-A will describe the assumptions and outline of our system model. Section III-B will describe our implementations for [4] in MIMO-OFDM. Section III-C will describe our implementation for [5] in MIMO-OFDM. Section III-D will combine both channel estimation and data detection and look at the system at the packet-level.

A. Systems Model

Consider a SVD MIMO-OFDM system shown in Figure 3 with R transmitters, Q receivers and N_c subcarriers.

For data symbols, if we denote $\mathbf{y}$ as the $QN_c \times 1$ received vector in the frequency domain where $\mathbf{y} = [\mathbf{y}(0)^T, \mathbf{y}(1)^T, \dots, \mathbf{y}(N_c - 1)^T]^T$ and $\mathbf{y}(a) = [y_1(a), \dots, y_Q(a)]^T$; $\mathbf{d}$ as the $RN_c \times 1$ transmitted vector in the frequency domain where $\mathbf{d} = [\mathbf{d}(0)^T, \mathbf{d}(1)^T, \dots, \mathbf{d}(N_c - 1)^T]^T$ and $\mathbf{d}(a) = [d_1(a), \dots, d_R(a)]^T$; $\mathbf{H}$ as

the $QN_c \times RN_c$ block diagonal matrix of the form $\begin{bmatrix} \mathbf{H}(0) & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \ddots & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{H}(N_c - 1) \end{bmatrix}$ where

$\mathbf{H}(n) = \begin{bmatrix} h_{1,1}(n) & \dots & h_{1,R}(n) \\ \vdots & \ddots & \vdots \\ h_{Q,1}(n) & \dots & h_{Q,R}(n) \end{bmatrix}$; $\mathbf{V}$ as the $RN_c \times KN_c$ pre-coding matrix $\begin{bmatrix} \mathbf{V}(0) & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \ddots & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{V}(N_c - 1) \end{bmatrix}$ where

$\mathbf{V}(n) = \begin{bmatrix} V_{1,1}(0) & \dots & V_{1,K}(0) \\ \vdots & \ddots & \vdots \\ V_{Q,1}(0) & \dots & V_{Q,K}(0) \end{bmatrix}$ where $K \leq R$ is the number of different data symbols per subcarrier;

$\mathbf{U}$ as the $KN_c \times QN_c$ de-coding matrix $\begin{bmatrix} \mathbf{U}(0) & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \ddots & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{U}(N_c - 1) \end{bmatrix}$ where $\mathbf{U}(n) = \begin{bmatrix} U_{1,1}(0) & \dots & U_{1,K}(0) \\ \vdots & \ddots & \vdots \\ U_{Q,1}(0) & \dots & U_{Q,K}(0) \end{bmatrix}$;

$\mathbf{n}$ be the $QN_c \times 1$ Gaussian noise vector with variance σ_n^2 ; $\mathbf{P}$ as the $N_c \times N_c$ diagonal phase-noise matrix where $\mathbf{P} = \text{diag}([e^{j\theta_0}, \dots, e^{j\theta_{N_c-1}}]^T)$; $\mathbf{E}$ as the $N_c \times N_c$ diagonal carrier frequency offset matrix where $\mathbf{E} = \text{diag}([1, \dots, e^{j2\pi\epsilon(N_c-1)/N_c}]^T)$; $\mathbf{F}$ and $\mathbf{F}^H$ are the DFT and IDFT matrices respectively; then we can rewrite (30) into

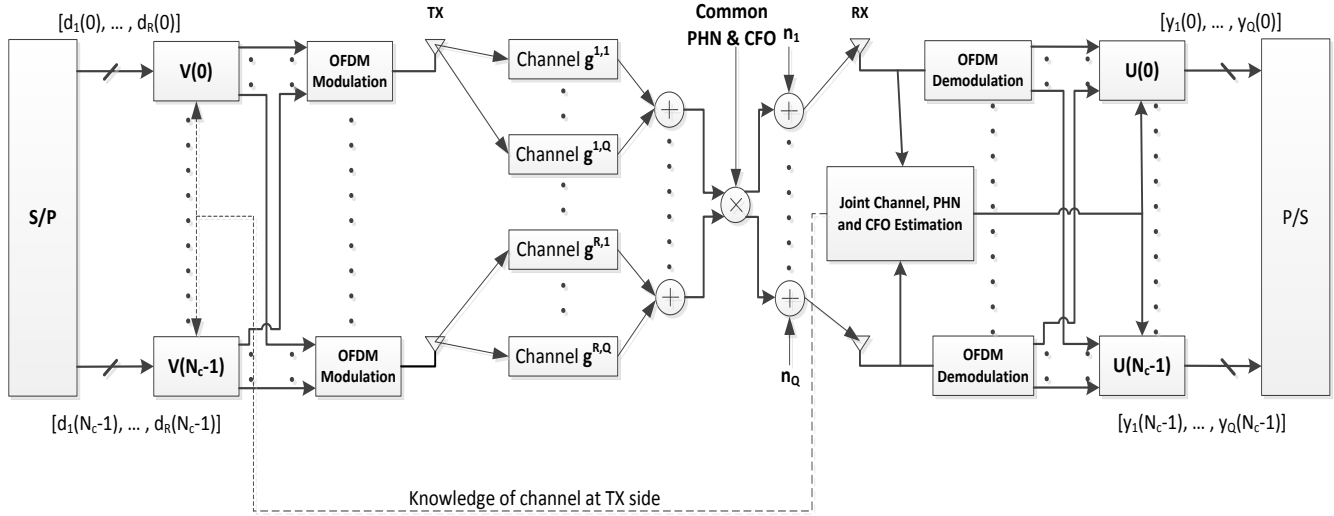


Figure 3. MIMO-OFDM System Model

$$\mathbf{y} = \mathbf{U}^H (\mathbf{F} \otimes \mathbf{I}_Q) [(\mathbf{E} \mathbf{P} \otimes \mathbf{I}_Q) (\mathbf{F}^H \otimes \mathbf{I}_Q) \mathbf{H} \mathbf{V} \mathbf{d} + \mathbf{n}] \quad (32)$$

where $\otimes$ represents the Kronecker product. We can make the assumption that the same phase noise sequence affects each receiver if we assume a common oscillator.

In the case of channel estimation where $\mathbf{d}$ is known but $\mathbf{H}$ is unknown, $\mathbf{U}$ and $\mathbf{V}$ would both be identity matrices. We can express the received $Q N_c \times 1$ time domain vector $\mathbf{r} = [\mathbf{r}_1, \dots, \mathbf{r}_Q]^T$ and $N_c \times 1$ vector $\mathbf{r}_n = [r_n(0), \dots, r_n(N_c - 1)]^T$ with

$$\mathbf{r}_n = \mathbf{E} \mathbf{P} \tilde{\mathbf{F}}^H \mathbf{A} \mathbf{g}_n + \mathbf{n} \quad (33)$$

$$\mathbf{r} = (\mathbf{I}_Q \otimes \mathbf{P})(\mathbf{I}_Q \otimes \mathbf{E})(\mathbf{I}_Q \otimes \tilde{\mathbf{F}}^H) \underline{\mathbf{A}} + \mathbf{n} \quad (34)$$

In the above $\tilde{\mathbf{F}} = [\mathbf{F}; \mathbf{F}]$ and $\mathbf{F}$ is $N_c/2 \times N_c/2$ FFT matrix. Let $\mathbf{W}$ be the $N_c/2 \times L$ matrix consisting of the first L columns and even rows of the $N_c \times N_c$ FFT matrix, then we can express $\mathbf{A}$ as a $N_c/2 \times RL$ matrix and $\underline{\mathbf{A}}$ as a $Q N_c/2 \times 1$ vector as

$$\mathbf{A} = [\text{diag}(\mathbf{D}_1) \mathbf{W} \quad \dots \quad \text{diag}(\mathbf{D}_R) \mathbf{W}] \quad \underline{\mathbf{A}} = \begin{bmatrix} \mathbf{A} \mathbf{g}_1 \\ \vdots \\ \mathbf{A} \mathbf{g}_Q \end{bmatrix} \quad (35)$$

where $\mathbf{D}_i = \text{diag}(\mathbf{d}_i)$ of the i^{th} transmitter and $\mathbf{g}_n$ is the $RL \times 1$ vector of the following form $\mathbf{g}_n = \begin{bmatrix} \mathbf{g}_{1n} \\ \vdots \\ \mathbf{g}_{Rn} \end{bmatrix}$ where $\mathbf{g}_{in} = \begin{bmatrix} g_{in}(0) \\ \vdots \\ g_{in}(L-1) \end{bmatrix}$.

In our model, we consider a slow fading frequency-selective channel where the CIR is assumed to remain constant during each packet of transmission, which consists of several OFDM symbols. An initial preamble is sent in order to achieve frequency synchronization and channel estimation. Then data symbols are sent, where frequency offset correction and channel equalization is performed using the estimates gained during the preamble stage. During both stages, the PHN is estimated and corrected for.

B. Joint MIMO-OFDM PHN and Data Estimation

First let us consider perfect frequency and channel estimation. This does not lose generality as frequency and channel estimation is performed with the preamble and is not re-estimated in the data portion of the packet. Once the preamble is implemented, we will simply use the estimated channel and CFO.

A simple way of adapting the algorithm (22) and (23) to the case of MIMO-OFDM is to consider them as MISO cases where at each receiver you receive the combined inputs of all the transmitters. If at the q th we let $\mathbf{r}_q$ be the time domain representation of $\mathbf{y}_q$ which can be expressed as

$$\mathbf{r}_q = \mathbf{P} \sum_{r=1}^R \mathbf{F}^H \mathbf{H}_{q,r} \tilde{\mathbf{V}} \mathbf{d} + \mathbf{n} \quad (36)$$

where $\tilde{\mathbf{V}}$ is the $(r+nR)$ th row of matrix $\mathbf{V}$ with $r \in 1, \dots, R$ and $n \in 0, \dots, N_c - 1$. We can see how equation (36) seem extremely similar to the system equation in (17) whose solution is provided by (22) and (23) with the exception that the SISO $\mathbf{F}^H \mathbf{H} \mathbf{d}$ term is replaced with the MISO $\sum_{r=1}^R \mathbf{F}^H \mathbf{H} \mathbf{V} \mathbf{d}$. However this is clearly suboptimal because it does not take advantage of the fact that the same phase noise sequence affects the inputs to each receiver. Further it cannot be generalized to the case where the phase noise sequence $\mathbf{P}$ is not the same at each receiver. In order to obtain a joint estimate based on all the receivers, we re-write equation (32) in the time domain where

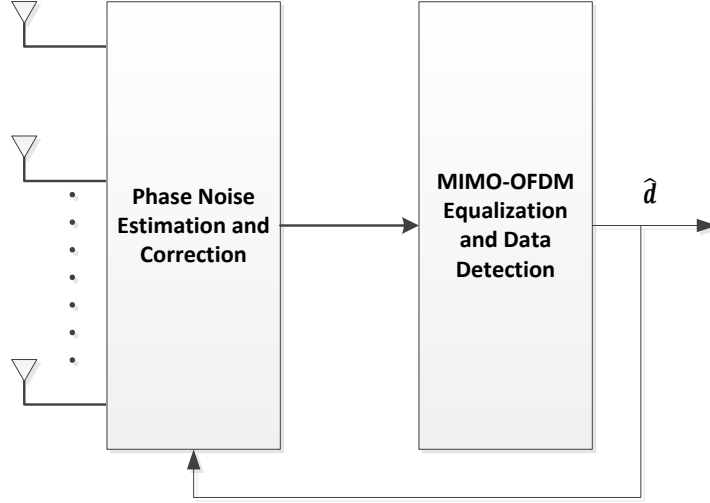


Figure 4. Iterative MIMO-OFDM Joint Estimation

$$\mathbf{r} = \left[\text{diag} \left((\mathbf{F}^H \otimes \mathbf{I}_Q) \mathbf{H} \mathbf{V} \mathbf{d} \right) (\mathbf{I}_{N_c} \otimes \mathbf{1}_Q) \mathbf{P} + \mathbf{n} \right] \quad (37)$$

In our case, the SISO term $\mathbf{F}^H \mathbf{H} \mathbf{d}$ is replaced by the MIMO term $\text{diag} \left((\mathbf{F}^H \otimes \mathbf{I}_Q) \mathbf{H} \mathbf{V} \mathbf{d} \right) (\mathbf{I}_{N_c} \otimes \mathbf{1}_Q)$. We can then rewrite (22) to

$$\hat{\boldsymbol{\theta}}^{(t)} = [\sigma_n^2 \boldsymbol{\Phi}^{-1} + \mathbf{X}^H \mathbf{X}]^{-1} \text{Im}[\mathbf{X}^H (\mathbf{r} - \mathbf{F}^H \mathbf{H} \hat{\mathbf{d}}^{(t-1)})] \quad (38)$$

where $\mathbf{X} = \text{diag}(\mathbf{F}^H \mathbf{H} \mathbf{V} \hat{\mathbf{d}}^{(t-1)}) (\mathbf{I}_{N_c} \otimes \mathbf{1}_Q)$. We can follow the steps in [4] to rigorously derive the MIMO equivalent of (23) given (38), however we notice that (23) is nothing more than PHN correction followed by MMSE equalization. Therefore we can simplify our calculations by implementing a MIMO-OFDM zero-forcing equalizer where we can write (23) as

$$\hat{\mathbf{d}}^{(t)} = \boldsymbol{\Sigma}^{-1} \mathbf{U}^H (\mathbf{F} \otimes \mathbf{I}_Q) (\mathbf{P}^H \otimes \mathbf{I}_Q) \mathbf{r} \quad (39)$$

where $\mathbf{P} = \text{diag}(e^{j\theta})$. A graphical interpretation of the iterative MIMO-OFDM equations for joint data and PHN estimation that we derived above is shown in Figure 4. The initial estimate is set to the estimate obtained by correcting for only the CPE. We also like to point out that the data estimate in (39)

must be quantized to the constellation set even though we assume a continuous prior distribution for data before using it for PHN estimate in equation (38). The full algorithm is summarized below in Figure 5.

Initialization:	$\hat{\mathbf{d}}^{(0)} = \hat{\mathbf{d}}_{CPE} \text{ and } \hat{\boldsymbol{\theta}}^{(0)} = \mathbf{0}$
Iterations:	for $i = 1$ to n iterations
Update $\hat{\boldsymbol{\theta}}$	$\hat{\boldsymbol{\theta}}^{(t)} = [\sigma_n^2 \boldsymbol{\Phi}^{-1} + \mathbf{X}^H \mathbf{X}]^{-1} \text{Im}[\mathbf{X}^H (\mathbf{r} - \mathbf{F}^H \mathbf{H} \hat{\mathbf{d}}^{(t-1)})]$
Update $\hat{\mathbf{d}}$	$\hat{\mathbf{d}}^{(t)} = \boldsymbol{\Sigma}^{-1} \mathbf{U}^H (\mathbf{F} \otimes \mathbf{I}_Q) (\mathbf{P}^H \otimes \mathbf{I}_Q) \mathbf{r}$
	end

Figure 5. Data and PHN Estimation

C. Joint MIMO-OFDM CFO, PHN and Channel Estimation

Training Symbol Sequence in MIMO-OFDM				
	TX 1	TX 2	...	TX R
0	$\pm\alpha \pm j\beta$	$\pm\alpha \pm j\beta$	...	$\pm\alpha \pm j\beta$
1	0	0	...	0
2	$\pm\alpha \pm j\beta$	$(\pm\alpha \pm j\beta)e^{\frac{-j2\pi L}{N_c/2}}$	...	$(\pm\alpha \pm j\beta)e^{\frac{-j2\pi(R-1)L}{N_c/2}}$
3	0	0	...	0
$\vdots$	$\vdots$	$\vdots$	...	$\vdots$
N_c-2	$\pm\alpha \pm j\beta$	$(\pm\alpha \pm j\beta)e^{\frac{-j2\pi L(N_c-1)/2}{N_c/2}}$	...	$(\pm\alpha \pm j\beta)e^{\frac{-j2\pi(R-1)L(N_c-1)/2}{N_c/2}}$
N_c-1	0	0	...	0

Figure 6. MIMO-OFDM Optimal Training Symbols

In order to adapt the work in II B.3, we need to first come up with a set of orthogonal pilot symbols that can be used for channel estimation. An optimal sequence was provided in (29) in B.5 and shown in Figure 6. As pointed out in B.3, the Moose algorithm used to estimate the CFO requires transmission of pilot sequences with two repetitive halves at the receiver. This can obviously be achieved in SISO cases by sending pilots only on even subcarriers and nulls on the odd subcarriers. If for the MIMO-OFDM case where we also transmit even powered symbols on even subcarriers and nulls on odd subcarriers, then we can write (8) in scalar form as

$$\begin{aligned}
 r_n^q &= \frac{1}{\sqrt{N_c/2}} e^{j(\theta_n + 2\pi\epsilon n/N_c)} \sum_{r=1}^R \sum_{k=0}^{N_c/2-1} h_k^{q,r} d_k^r e^{-j4\pi(r-1)Lk/N_c} e^{j4\pi nk/N_c} + n_n \\
 &= \frac{1}{\sqrt{N_c/2}} e^{j(\theta_n + 2\pi\epsilon n/N_c)} \sum_{k=0}^{N_c/2-1} (\sum_{r=1}^R h_k^{q,r} d_k^r e^{-\frac{j4\pi(r-1)Lk}{N_c}}) e^{j4\pi nk/N_c} + n_n
 \end{aligned} \tag{40}$$

As can be seen in (40), the combined signal at each receiver q also consists of two repetitive halves. Therefore transmission of the optimal pilot symbols that satisfy the conditions set in II B.5 will also satisfy the conditions for estimating the CFO using the Moose algorithm. Further this means that as long as $P > LR$ is satisfied, we can estimate all the QR channels with one OFDM symbol. Also, because we are only transmitting on half of the subcarriers, each pilot symbol can be transmitted with double the power as before.

C.1 CFO Estimation

We begin similarly to equation (9) in section II B.2 where we have

$$\mathbf{r}_1 = (\mathbf{P}_1 \otimes \mathbf{I}_{Q/2}) \mathbf{x} + \mathbf{n}_1 \quad \text{and} \quad \mathbf{r}_2 = e^{j\pi\epsilon} (\mathbf{P}_2 \otimes \mathbf{I}_{Q/2}) \mathbf{x} + \mathbf{n}_2 \tag{41}$$

However, in our MIMO-OFDM case, we have $QN_c/2 \times 1$ vectors $\mathbf{r}_1 = [\mathbf{r}(0)^T, \dots, \mathbf{r}(\frac{N_c}{2} - 1)^T]^T$ and $\mathbf{r}_2 = [\mathbf{r}(N_c/2)^T, \dots, \mathbf{r}(N_c - 1)^T]^T$, where $\mathbf{r}(n) = [r_1(n), \dots, r_Q(n)]^T$. Further $\mathbf{x} = (\mathbf{E}_1 \otimes \mathbf{I}_Q)(\tilde{\mathbf{F}}^H \otimes \mathbf{I}_Q)$, where $\mathbf{E}_1 = \text{diag}\left(\left[1, e^{\frac{j2\pi\epsilon}{N_c}}, \dots, e^{\frac{j2\pi(N_c/2-1)\epsilon}{N_c}}\right]^T\right)$ is $QN_c/2 \times QN_c/2$ matrix consisting of the first half of the carrier frequency offset sequence, $\tilde{\mathbf{F}} = [\mathbf{F}; \mathbf{F}]$ where $\mathbf{F}$ is $N_c/2 \times N_c/2$ FFT matrix, $\mathbf{H} = \begin{bmatrix} \mathbf{H}(0) & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \ddots & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{H}(N_c - 2) \end{bmatrix}$ is $QN_c/2 \times RN_c/2$ matrix where $\mathbf{H}(n) = \begin{bmatrix} H_{11}(n) & \dots & H_{1R}(n) \\ \vdots & \ddots & \vdots \\ H_{Q1}(n) & \dots & H_{QR}(n) \end{bmatrix}$ and $\mathbf{d} = [\mathbf{d}(0)^T, \dots, \mathbf{d}(N_c - 2)^T]^T$ where $\mathbf{d}(n) = [d_1(n), \dots, d_R(n)]^T$.

Similar to II B.2 equation (10) we combine the two equations in (41) to obtain

$$\mathbf{r}_2 = e^{j\pi\epsilon} (\mathbf{P}_\Delta \otimes \mathbf{I}_{Q/2}) \mathbf{r}_1 - e^{j\pi\epsilon} (\mathbf{P}_\Delta \otimes \mathbf{I}_{Q/2}) \mathbf{n}_1 + \mathbf{n}_2 \tag{42}$$

We can rewrite (42) in the following manner

$$\mathbf{r}_2 = e^{j\pi\epsilon} \text{diag}(\mathbf{r}_1) (\mathbf{I}_{N_c/2} \otimes \mathbf{1}_Q) \mathbf{P}_\Delta - e^{j\pi\epsilon} (\mathbf{P}_\Delta \otimes \mathbf{I}_{Q/2}) \mathbf{n}_1 + \mathbf{n}_2 \quad (43)$$

By apply the same steps as Lemma 2 of [5], it is apparent that the estimate for $\hat{\epsilon}$ in Section II B.2 equation (12) now becomes

$$\hat{\epsilon} = \frac{1}{\pi} \Re \left\{ \mathbf{r}_1^H (\mathbf{R}_1 (\mathbf{I}_{N_c/2} \otimes \mathbf{1}_Q) \Phi_\Delta (\mathbf{I}_{N_c/2} \otimes \mathbf{1}_Q)^H \mathbf{R}_1^H + 4\sigma_n^2 \mathbf{I})^{-1} \mathbf{r}_2 \right\} \quad (44)$$

From $\hat{\epsilon}$, we can construct the $\hat{\mathbf{E}}$ matrix.

C.2 Channel Estimation

Using equation (33), we can see that each receiver can find the channel between all pairings between itself and any transmitters through a simple LS estimate

$$\hat{\mathbf{g}}_n = \frac{1}{4\rho^2} \mathbf{A}^H \tilde{\mathbf{F}} \mathbf{P}^H \mathbf{E}^H \mathbf{r}_n \quad (45)$$

where $\mathbf{g}_n = [\mathbf{g}_{1n}(\mathbf{0})^T, \dots, \mathbf{g}_{Rn}(\mathbf{0})^T]^T$ where $\mathbf{g}_{in} = [g_{in}(0), \dots, g_{in}(L-1)]^T$. Notice that by transforming the channel estimate into the time domain where the number of channel taps L is less than the number of channel taps N_c in the frequency domain, we are able to lower the number of distinct pilot symbols that is required to uniquely determine the channel.

C.3 Phase Estimation

We can re-express equation (34) through substitution of equation (45) in matrix form

$$\mathbf{r} = \begin{bmatrix} \mathbf{P} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \ddots & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{P} \end{bmatrix} \begin{bmatrix} \mathbf{E} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \ddots & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{E} \end{bmatrix} \begin{bmatrix} \tilde{\mathbf{F}}^H & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \ddots & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \tilde{\mathbf{F}}^H \end{bmatrix} \begin{bmatrix} \frac{1}{4\rho^2} \mathbf{A} \mathbf{A}^H \tilde{\mathbf{F}} \mathbf{P}^H \mathbf{E}^H \mathbf{r}_1 \\ \vdots \\ \frac{1}{4\rho^2} \mathbf{A} \mathbf{A}^H \tilde{\mathbf{F}} \mathbf{P}^H \mathbf{E}^H \mathbf{r}_Q \end{bmatrix} + \mathbf{n}$$

$$= \begin{bmatrix} \frac{1}{4\rho^2} \mathbf{P} \mathbf{E} \tilde{\mathbf{F}}^H \mathbf{A} \mathbf{A}^H \tilde{\mathbf{F}} \mathbf{P}^H \mathbf{E}^H \mathbf{r}_1 \\ \vdots \\ \frac{1}{4\rho^2} \mathbf{P} \mathbf{E} \tilde{\mathbf{F}}^H \mathbf{A} \mathbf{A}^H \tilde{\mathbf{F}} \mathbf{P}^H \mathbf{E}^H \mathbf{r}_Q \end{bmatrix} + \mathbf{n} \quad (46)$$

Since the likelihood function can be expressed as

$$\begin{aligned} L(\theta) &= \frac{1}{2\sigma^2} \left(\mathbf{r} - \begin{bmatrix} \frac{1}{4\rho^2} \mathbf{P} \mathbf{E} \tilde{\mathbf{F}}^H \mathbf{A} \mathbf{A}^H \tilde{\mathbf{F}} \mathbf{P}^H \mathbf{E}^H \mathbf{r}_1 \\ \vdots \\ \frac{1}{4\rho^2} \mathbf{P} \mathbf{E} \tilde{\mathbf{F}}^H \mathbf{A} \mathbf{A}^H \tilde{\mathbf{F}} \mathbf{P}^H \mathbf{E}^H \mathbf{r}_Q \end{bmatrix} \right)^H \left(\mathbf{r} - \begin{bmatrix} \frac{1}{4\rho^2} \mathbf{P} \mathbf{E} \tilde{\mathbf{F}}^H \mathbf{A} \mathbf{A}^H \tilde{\mathbf{F}} \mathbf{P}^H \mathbf{E}^H \mathbf{r}_1 \\ \vdots \\ \frac{1}{4\rho^2} \mathbf{P} \mathbf{E} \tilde{\mathbf{F}}^H \mathbf{A} \mathbf{A}^H \tilde{\mathbf{F}} \mathbf{P}^H \mathbf{E}^H \mathbf{r}_Q \end{bmatrix} \right) + \frac{1}{2} \theta^T \Phi^{-1} \theta \\ &= \frac{1}{2\sigma^2} \left(\mathbf{r}^H \mathbf{r} - \mathbf{r}^H \begin{bmatrix} \frac{1}{4\rho^2} \mathbf{P} \mathbf{E} \tilde{\mathbf{F}}^H \mathbf{A} \mathbf{A}^H \tilde{\mathbf{F}} \mathbf{P}^H \mathbf{E}^H \mathbf{r}_1 \\ \vdots \\ \frac{1}{4\rho^2} \mathbf{P} \mathbf{E} \tilde{\mathbf{F}}^H \mathbf{A} \mathbf{A}^H \tilde{\mathbf{F}} \mathbf{P}^H \mathbf{E}^H \mathbf{r}_Q \end{bmatrix} \right) + \frac{1}{2} \theta^T \Phi^{-1} \theta \\ &= \frac{1}{2\sigma_n^2} \left(\sum_{n=1}^Q \mathbf{r}_n^H \mathbf{r}_n - \frac{1}{4\rho^2} \mathbf{r}_n^H \mathbf{P} \mathbf{E} \tilde{\mathbf{F}}^H \mathbf{A} \mathbf{A}^H \tilde{\mathbf{F}} \mathbf{P}^H \mathbf{E}^H \mathbf{r}_n \right) + \frac{1}{2} \theta^T \Phi^{-1} \theta \\ &= \frac{1}{2\sigma_n^2} \mathbf{u}^T \mathbf{E} \left(\sum_{n=1}^Q (4\rho^2 \mathbf{R}_n^H \mathbf{R}_n - \mathbf{R}_n^H \tilde{\mathbf{F}}^H \mathbf{A} \mathbf{A}^H \tilde{\mathbf{F}} \mathbf{R}_n) \right) \mathbf{E}^H \mathbf{u}^* + \frac{1}{2} \theta^T \Phi^{-1} \theta \end{aligned} \quad (47)$$

where $\mathbf{R}_n = \text{diag}(\mathbf{r}_n)$ and $\mathbf{u} = [e^{j\theta_0}, e^{j\theta_1}, \dots, e^{j\theta_{N_c-1}}]$. The above equation is similar to equation (14) in Section II B.2 except in the MIMO-OFDM case, the term $\mathbf{A}$ is equal to $\sum_{n=1}^Q (4\rho^2 \mathbf{R}_n^H \mathbf{R}_n - \mathbf{R}_n^H \tilde{\mathbf{F}}^H \mathbf{A} \mathbf{A}^H \tilde{\mathbf{F}} \mathbf{R}_n)$. Therefore the optimal phase noise estimate becomes

$$\begin{aligned} \hat{\theta} &= [\text{Re}(\hat{\mathbf{E}}(\mathbf{A}) \hat{\mathbf{E}}^H) + 4\rho^2 \sigma_n^2 \Phi^{-1}]^{-1} \text{Im}(\hat{\mathbf{E}}(\mathbf{A}) \hat{\mathbf{E}}^H) \mathbf{1}_{N_c} \\ \text{where, } \mathbf{A} &= \sum_{n=1}^Q (4\rho^2 \mathbf{R}_n^H \mathbf{R}_n - \mathbf{R}_n^H \tilde{\mathbf{F}}^H \mathbf{A} \mathbf{A}^H \tilde{\mathbf{F}} \mathbf{R}_n) \end{aligned} \quad (48)$$

In summary, we derive the following set of closed-form equations that can be used to estimate a channel affected by CFO and PHN. The full outline of the steps is shown in Fig. 7.s

$$\begin{aligned}
\text{Step 1: } \hat{\epsilon} &= \frac{1}{\pi} \nabla r_1^H (R_1 (\mathbf{I}_{N_c/2} \otimes \mathbf{1}_Q) \Phi_\Delta (\mathbf{I}_{N_c/2} \otimes \mathbf{1}_Q)^H R_1^H + 4\sigma_n^2 I)^{-1} r_2 \\
\hat{E} &= \text{diag}([1, e^{j2\pi\hat{\epsilon}/N_c}, \dots, e^{j2\pi(N_c-1)\hat{\epsilon}/N_c}]) \\
\text{Step 2: } \hat{\theta} &= [\text{Re}(\hat{E}(A)\hat{E}^H) + 4\rho^2\sigma_n^2\Phi^{-1}]^{-1} \text{Im}(\hat{E}(A)\hat{E}^H) \mathbf{1}_{N_c} \\
P &= \text{diag}([e^{j\hat{\theta}_0}, \dots, e^{j\hat{\theta}_{N_c-1}}]) \\
\text{Step 3: } \hat{g}_n &= \frac{1}{4\rho^2} A^H \tilde{F} P^H E^H r_n \quad \text{for each receiver } 1 \dots Q
\end{aligned}$$

Figure 7. Joint CFO, PHN and Channel Estimation

D. Combined Packet Structure

Consider a more complete system with data symbols following the preamble in a structure shown in Figure 1. There are two areas that we need to address when we extend our model to a packet: 1) covariance matrix of PHN which would keep growing with each OFDM symbol in the packet and 2) energy allocation of each symbol as a percentage of total packet energy.

First, let us assume that the oscillator is perfect synchronized at the beginning of each packet. As synchronization only occurs at the beginning of a packet, this means the n th OFDM symbol within a packet will experience an $N_c \times N_c$ covariance matrix

$$\Phi(n) = \alpha_\varphi^2 \begin{bmatrix} (n-1)N_c + 1 & (n-1)N_c + 1 & \dots & (n-1)N_c + 1 \\ (n-1)N_c + 1 & (n-1)N_c + 2 & \dots & (n-1)N_c + 2 \\ \vdots & \vdots & \ddots & \vdots \\ (n-1)N_c + 1 & (n-1)N_c + 2 & \dots & (n-1)N_c + N_c \end{bmatrix} \quad (41)$$

Second, consider a system where a total energy of ψ is available per packet and each packet consists of g number of OFDM symbols including the preamble. In normal energy allocation, each OFDM symbol in the packet will receive $1/g$ percent of the total energy ψ . Therefore if our packet consist of one OFDM preamble symbol and $g-1$ data symbols, we say that in order to increase the energy allocated to the preamble by x percent more than its allocation of $1/g$ percent of the total energy ψ , then we must lower the energy allocated to each data symbol by $\frac{x}{g-1}$ percent. Vice versa if we want to decrease the energy allocated to the preamble by x percent less than $1/g$ percent of the total energy ψ , then we can increase the energy allocated to each data symbol by $\frac{x}{g-1}$ percent.

IV. Results

THE PROPOSED methods for joint estimation were simulated on a system with the following specifications for each stage described in Section III-B and shown in Figure 8. In the simulation, Wiener PHN was generated using Cholesky Decomposition. The CFO ϵ is a randomly selected value between $[-0.4, 0.4]$. The channel impulse response is an L -tapped independent complex Gaussian distributed with exponential decay. As specified in [25], the channel coherence time for LTE system is about 5 OFDM symbols which we will use to form a packet structure of 1 preamble followed by 4 data symbols.

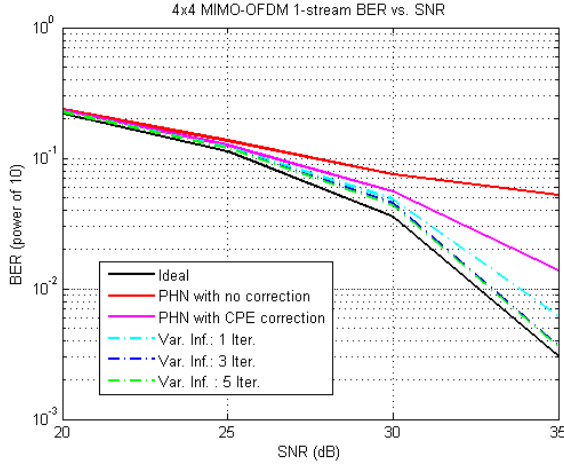
MIMO-OFDM Simulation Specifications	
Transmitters (R)	4
Receivers (Q)	4
Sampling Rate (Fs)	20 MHz
Subcarriers (N_c)	128
CP (N_g)	12
Taps (L)	10
Modulation	64-QAM

Figure 8. MIMO-OFDM Specifications

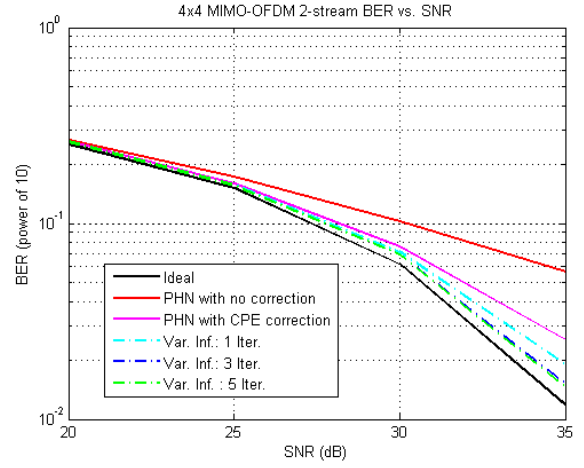
Joint Data and PHN Estimation: We used subcarriers 26, 40, 54, 68, 82, 96, 110, 126 for pilot carriers in the data symbol to estimate the common phase error. Figures 9, 10, and 11 show the results for 1° of Wiener PHN.

In Figure 9(a), (b), (c), we see that the performance of our proposed algorithm outperforms the conventional method of removing only the CPE. From Figure 10 and 11, we find that our algorithm tracks the PHN as well in the MIMO-OFDM case as the original algorithm did in SISO-OFDM (bear in mind that since there are 128 or $\sim 10^2$ subcarriers, the MSE of the estimate is larger on aggregate, but the same on a per PHN point basis). Further we notice that when sending one and two data-streams, PHN errors form the limiting case for BER for our proposed method. When four independent data streams are sent, the BER is limited by the SNR and interference cause by noise which results in all the curves being extremely close. We can further see that the phase noise estimate is not affected by the interference between independent data streams between transmitters by looking at the PHN MSE in Figure 10(a), (b),

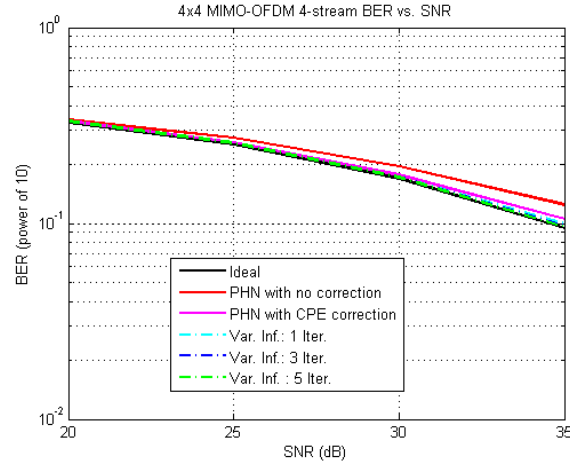
and (c) where the MSE vs. SNR curves are similar and Figure 11(a), (b), and (c) where we see that the PHN tracking performance is equivalent for different number of data streams. As in the SISO-OFDM case, we observe that the algorithm converges rapidly with no further improvement observed after 5 iterations.



(a)

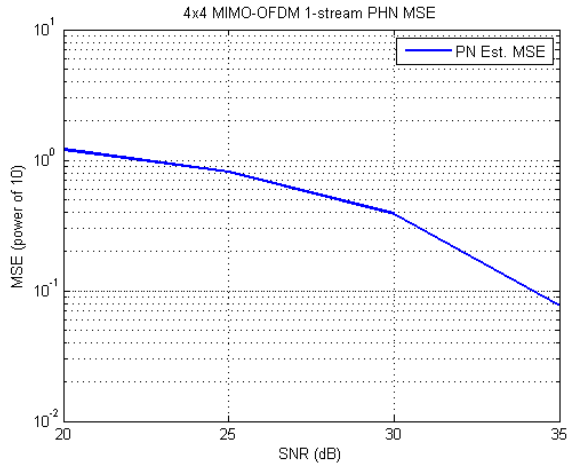


(b)

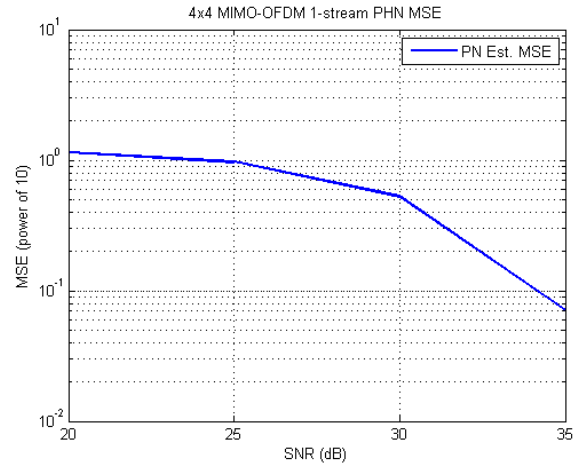


(c)

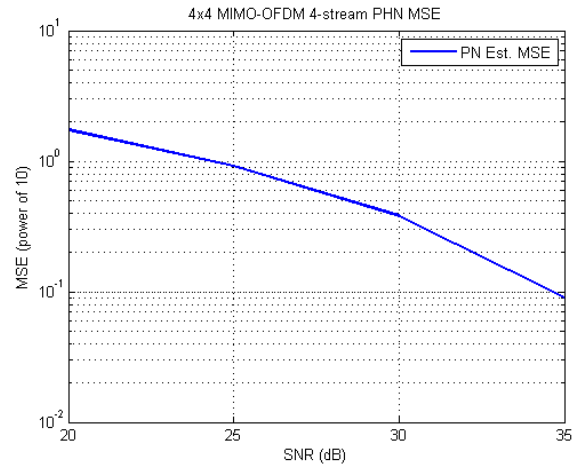
Figure 9. BER performance comparison between conventional method and proposed detection scheme for 1(a),2(b),4(c) independent data streams in 4x4 MIMO-OFDM system



(a)



(b)



(c)

Figure 10. PHN MSE for 1(a),2(b),4(c) independent streams are similar

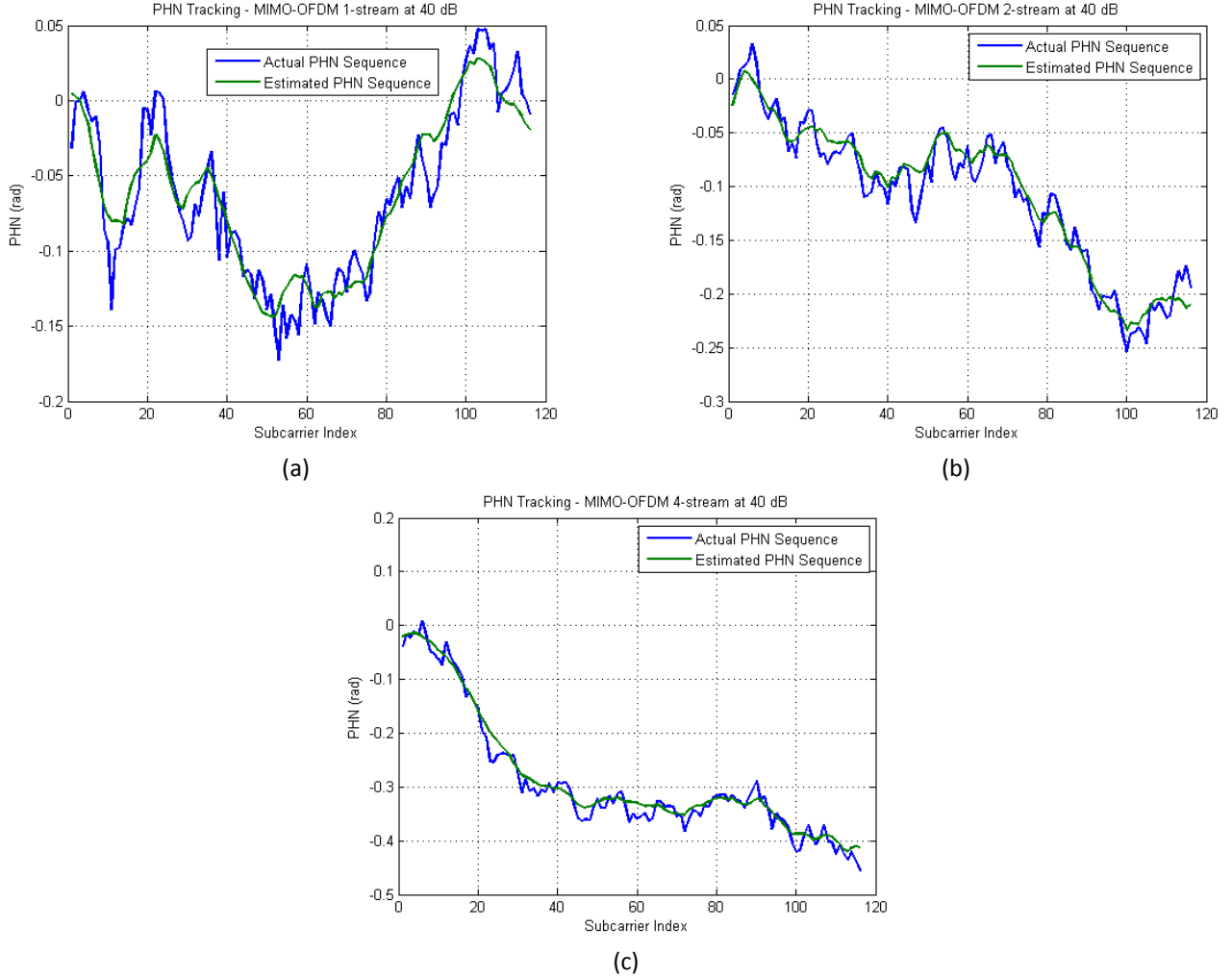


Figure 11. PHN Tracking for 1(a), 2(b), 4(c) independent streams show equivalent performance

Joint PHN, CFO and Channel Estimation: In Figure 12, we show the MSE vs. SNR for channel estimation for Wiener PHN. Although it is apparent that the proposed method outperforms methods that consider only CFO, two interesting points were observed that were absent from literature: 1) we observed the existence of an error floor in the CFO estimate that were dependent on the level of PHN (Figure 13) and 2) the correction of the RCPR (residual common phase rotation) was detrimental and not required. This is in contrast to the observed result in [5] where an error floor was not observed and the correction for RCPR (a constant difference represented as the mean of the difference between estimated sequence and actual sequence) was required in order to fully correct for the PHN sequence.

It is yet unknown why there is this discrepancy in results. We were able to remove the error floor in Figure 14 by removing the PHN during the CFO estimate. We are unsure how this error floor was

overcome in [5]. We believe that the second discrepancy is related to the existence of the error floor as we observed that we indeed need to remove the RCPR in Figure 14 where the error floor does not exist.

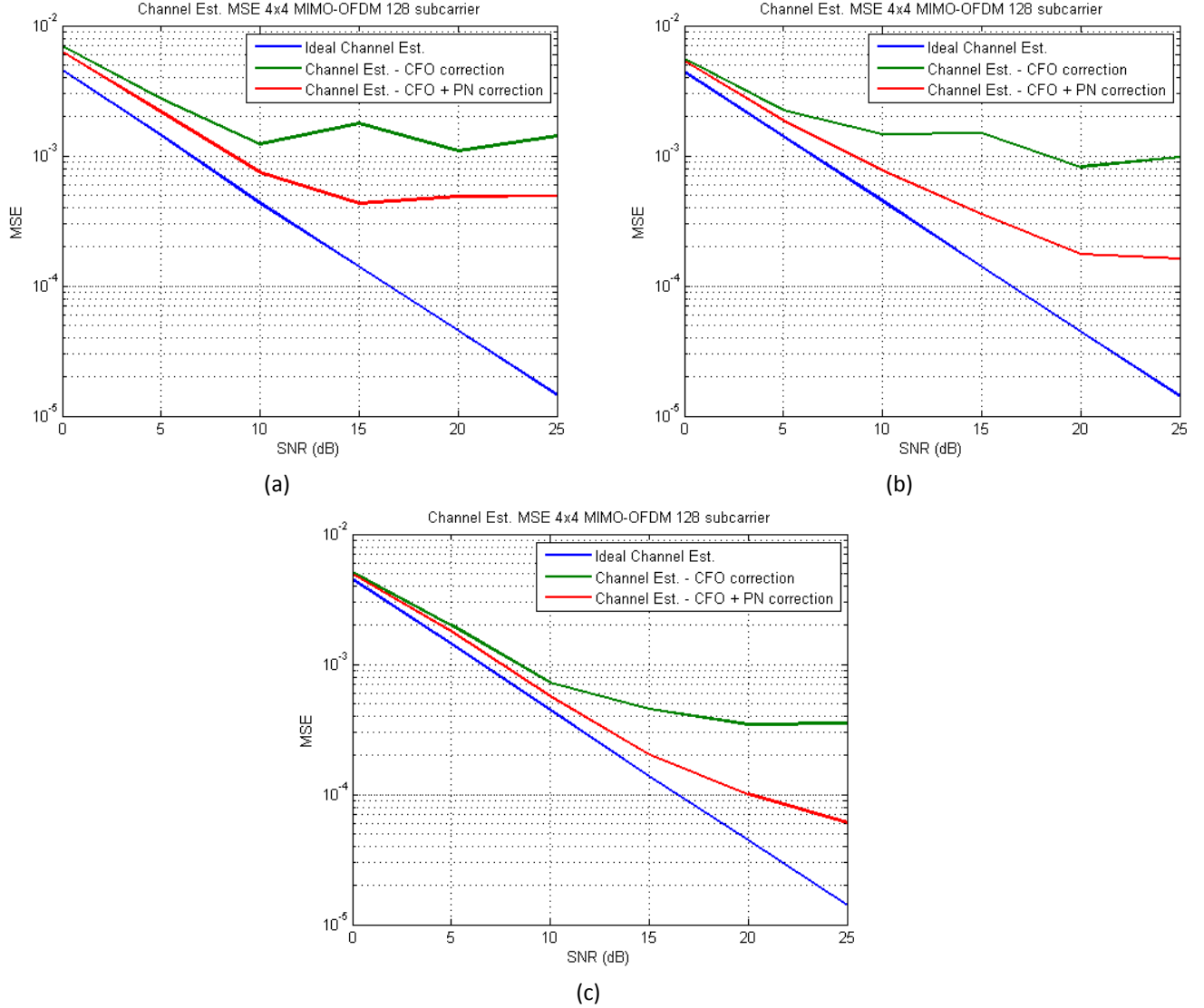


Figure 12. MSE comparisons of proposed joint estimation method and CFO correction only for PHN of 1° (a), 0.75° (b), 0.5° (c)

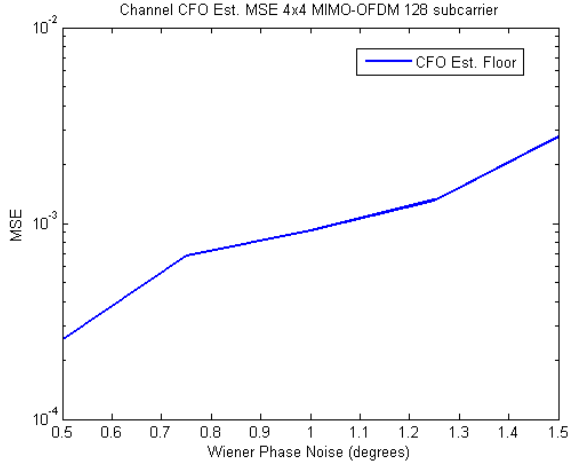


Figure 13. CFO estimate floor

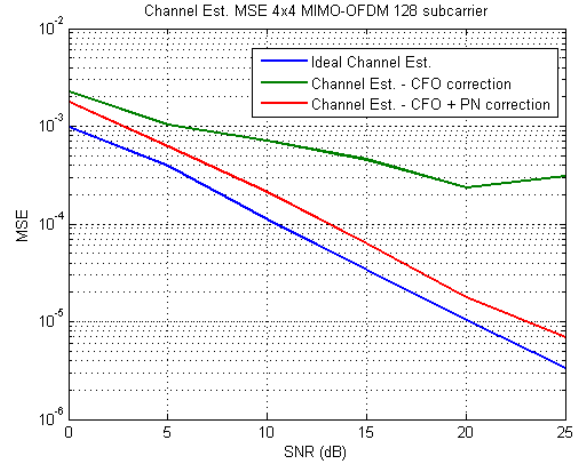


Figure 14. MSE with PHN removed during CFO estimate

Combined Packet Structure: We combined the above two methods into packet structure consisting of one preamble for channel estimation followed by 4 data packets. For data detection we used our variation inference algorithm with 5 iterations. For channel estimation, we used our MLE estimator. In our implementation, Wiener phase noise was studied.

The results are summarized in Fig. 15 and 16 where 0.5° , 0.75° , 1° of phase noise were added to the system. We observe that the optimal energy allocation is not monotonic and changes depending on the level of phase noise. However we observe in general two regimes of behavior: less than SNR of 30 dB and greater than SNR of 30 dB. Up to a threshold in SNR, which in our case is 30 dB, it appears that allocating 50% of the energy away from the preamble is always optimal and in general allocating energy away from the preamble outperforms allocating energy away from data symbols. However, past 30 dB, we observe this heuristic to be reversed. This effect is more pronounced with higher levels of PHN and can be observed more clearly in Figure 16(b) and (c). In all three cases at SNR greater than 30 dB, there is an energy allocation that clearly becomes the optimal: 0% (even) for 0.5° , 25% for 1° and 75% for 1.5° of PHN respectively.

While currently the reason for the two regimes of behavior is not exactly understood, we believe that it is related to the error floor caused by the CFO estimate that we observed during the channel estimation stage. However, this theory seems counterintuitive, as the existence of an error floor at high SNR would indicate allocating energy to the preamble at lower SNR and away from the preamble at high SNR. This is contrary to our data. Further work should be carried out to plot the MSE of the channel along with the BER of the data to fully ascertain some causal relationship, which should be

trivial addition to our model. However due to time constraints (>6 hours per simulation), we are unable to accomplish as of the writing of this final thesis report.

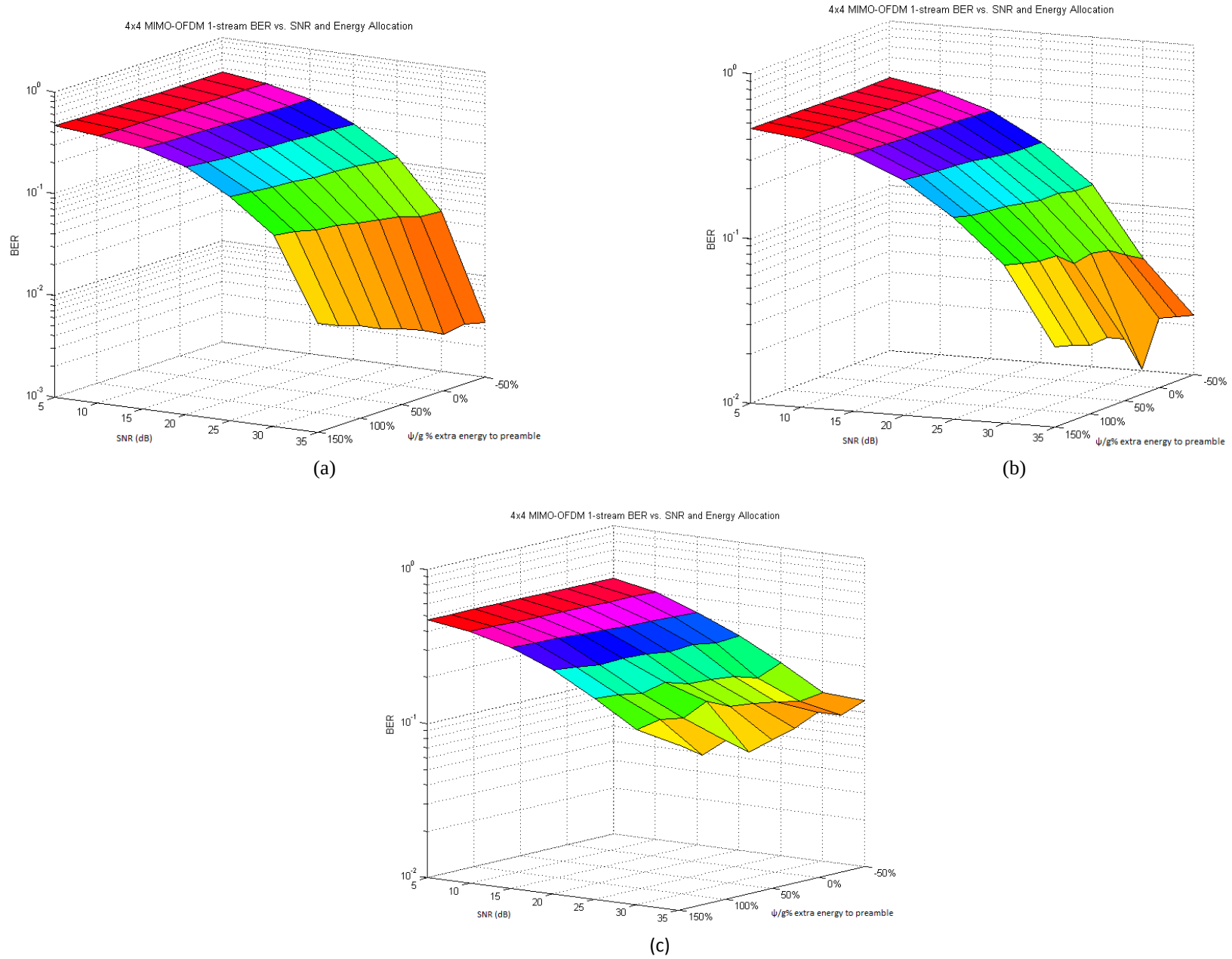
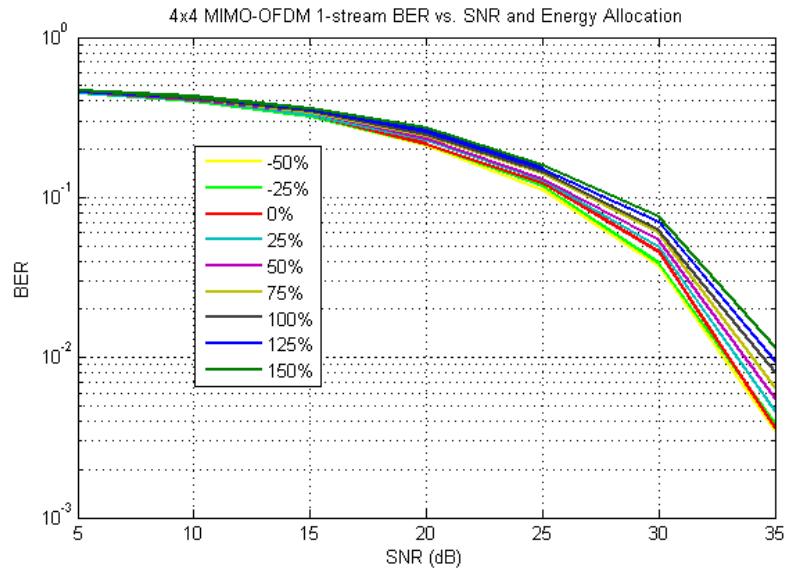
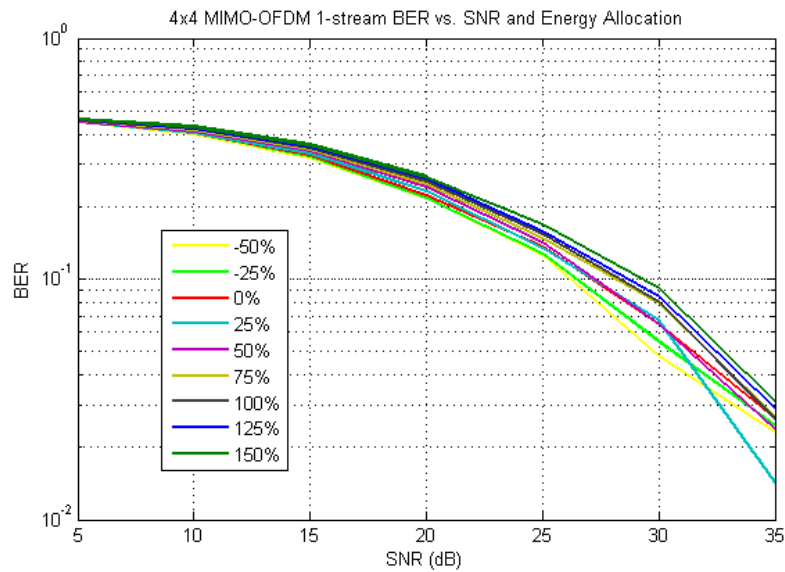


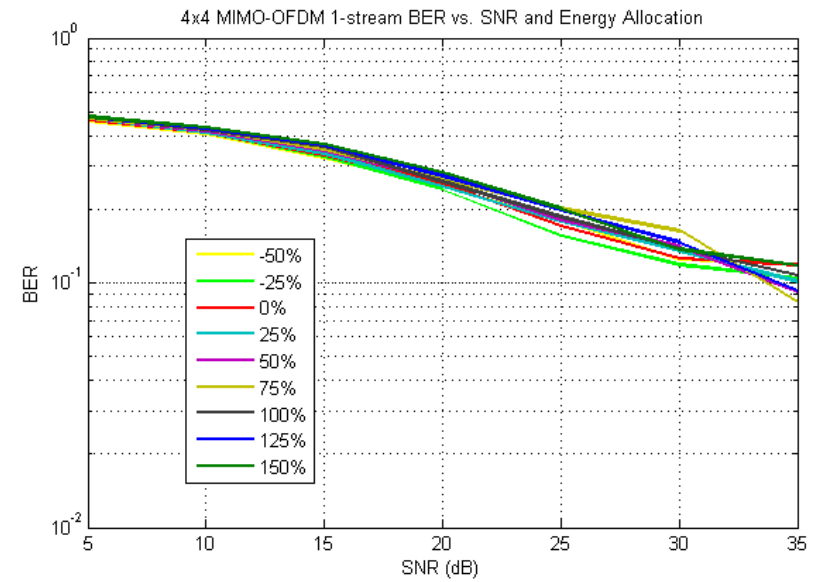
Figure 155. 3D view – Optimal energy allocation varies depending on phase noise: 0.5° (a), 0.75° (b), 1° (c) Wiener PHN



(a)



(b)



(c)

Figure 166. Side view – Improvement allocating energy away from preamble for SNR < 30 dB; Improvement allocating energy to preamble for SNR > 30 dB

V. Conclusion

THIS THESIS represents a complete physical layer overview of MIMO-OFDM receivers (channel estimation and data detection) in the presence of CFO and PHN. We were able to successfully extend the Variational Inference algorithm used for SISO-OFDM to the case of MIMO-OFDM. Further we also were able to extend the MLE for joint CFO, PHN and channel estimate from SISO-OFDM to MIMO-OFDM. Although discrepancy was observed with the original SISO-OFDM case in [4] and [5], the results nevertheless still show significant improvement over original schemes. Further, this thesis also explored basic energy allocation between preamble and data symbols and characterized the data error rates under CFO, PHN and imperfect channel.

In the future, we hope to resolve the discrepancy in results in our model and the model observed in literature for the SISO-OFDM case in [4] and [5]. Further research can be done in extending this thesis to the case of multi-user. Finally reduction in computational complexity can be achieved and implemented using the gradient descent algorithm as described in [4] and [5], which would provide a more elegant and implementable solution.

VII. References

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